ROOT CAUSE ANALYSIS OF U.S. AVIATION BOTTLENECKS USING PYTHON (2020 - 2026)

1. Project Overview

This project focuses on analysing flight delay data to understand patterns and trends in airline operations. The analysis aims to identify the major causes of flight delays, compare airport and airline performance, understand monthly and yearly delay patterns, and examine the relationship between delayed, cancelled, and diverted flights.

The project combines Data Cleaning, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis, and Data Visualization to transform raw flight data into meaningful insights.

The findings help identify operational challenges and provide data-driven recommendations to improve airline efficiency and reduce flight disruptions.

2. Dataset Information

Attribute	Details
Data Source	U.S. Bureau of Transportation Statistics (BTS)
Location	United States
Domain	Aviation / Transportation
Period	2020–2026
Original Records	28,729
Records After Cleaning	28,718
Original Attributes	21
Final Attributes	30
File Format	CSV
Environment	Google Colab

The dataset was obtained from the U.S. Bureau of Transportation Statistics (BTS). It contains information about U.S. airline operations, including arriving flights, delayed flights, cancellations, diversions, and different causes of flight delays.

The dataset also includes information about airlines, airports, years, and months, which helps analyse flight delay patterns and compare airline and airport performance.

3. Problem Statement

Flight delays are a common challenge in the airline industry. They can affect passenger satisfaction, airline operations, flight schedules, and overall operational efficiency. Delays may occur due to different factors, such as carrier-related issues, weather conditions, National Aviation System (NAS) issues, security problems, and late-arriving aircraft.

Understanding these delays is important for identifying operational problems and improving airline performance. However, analysing flight data across multiple years, months, airports, airlines, and delay causes can be challenging without proper data analysis.

This project aims to analyze flight delay data to identify patterns, trends, and major causes of delays. It also focuses on comparing airline and airport performance to generate meaningful insights into flight operations.

4. Objectives

The specific objectives are:

1. To analyse flight delay trends across different years and months.
2. To compare flight delays and delay rates across airports and airlines.
3. To identify the major causes of flight delays.
4. To analyse the relationship between flight delays, cancellations, and diversions.
5. To compare total flight delay duration across different causes.

5. Attribute (Column / Feature)

The final dataset contains 30 attributes, including 21 original attributes and 9 additional features created through feature engineering.

Attribute Name	Description
year	Year in which the flights operated
month	Month in which the flights operated
carrier	Unique code representing the airline
carrier_name	Full name of the airline
airport	Code representing the airport
airport_name	Full name of the airport
arr_flights	Total number of arriving flights
arr_del15	Number of arriving flights delayed by 15 minutes or more
carrier_ct	Number of delayed flights due to airline-related issues
weather_ct	Number of delayed flights due to weather conditions
nas_ct	Number of delayed flights due to National Aviation System (NAS) issues
security_ct	Number of delayed flights due to security-related issues
late_aircraft_ct	Number of delayed flights due to late-arriving aircraft
arr_cancelled	Total number of cancelled arriving flights
arr_diverted	Total number of diverted arriving flights

arr_delay	Total arrival delay duration in minutes
carrier_delay	Total delay duration caused by airline-related issues
weather_delay	Total delay duration caused by weather conditions
nas_delay	Total delay duration caused by National Aviation System (NAS) issues
security_delay	Total delay duration caused by security-related issues
late_aircraft_delay	Total delay duration caused by late-arriving aircraft

Feature Engineering columns (9 Columns)

Attribute Name	Description
delay_rate	Percentage of arriving flights delayed by 15 minutes or more
cancellation_rate	Percentage of arriving flights that were cancelled
avg_delay_duration	Average delay duration per delayed flight
carrier_share	Percentage contribution of carrier-related issues to overall flight delays
weather_share	Percentage contribution of weather-related issues to overall flight delays
nas_share	Percentage contribution of National Aviation System issues to overall flight delays
security_share	Percentage contribution of security-related issues to overall flight delays
late_aircraft_share	Percentage contribution of late-arriving aircraft to overall flight delays
season	Season categorized based on the month of flight operation

The additional features were created to make comparisons easier, understand the contribution of different delay causes, and identify seasonal patterns in flight delays.

6. Tools & Technologies

- Python
- Pandas – Data manipulation and analysis
- NumPy – Numerical operations
- Matplotlib – Data visualization
- Seaborn – Statistical visualization
- Google Colab

7. Data Pre-Processing and Feature Engineering

The dataset was inspected and cleaned before performing further analysis to ensure data quality and consistency.

The following validations were performed:

- Checked the data types of all columns.
- Verified that no duplicate records were present.
- Checked numerical columns for negative values. No negative values were found.
- Validated the `month` column to ensure values were between 1 and 12.
- Confirmed that the missing values in `arr_del15` were handled.

Missing Value Handling

- The `arr_flights` column contained 11 missing values. These records were removed.
- The `arr_del15` column initially contained 48 missing values.
- After removing the 11 records, 37 missing values remained in `arr_del15`.
- These 37 missing values were replaced with 0 after checking the corresponding flight records.

Feature Engineering

Feature engineering was performed to create additional columns that support better analysis and comparisons.

The following 9 features were created:

Feature	Description
`delay_rate`	Percentage of arriving flights delayed by 15 minutes or more.
`cancellation_rate`	Percentage of arriving flights that were cancelled.
`avg_delay_duration`	Average delay duration per delayed flight.
`carrier_share`	Percentage contribution of carrier-related issues to total arrival delay minutes.
`weather_share`	Percentage contribution of weather conditions to total arrival delay minutes.
`nas_share`	Percentage contribution of National Aviation System issues to total arrival delay minutes.
`security_share`	Percentage contribution of security-related issues to total arrival delay minutes.
`late_aircraft_share`	Percentage contribution of late-arriving aircraft to total arrival delay minutes.
`season`	Categorizes months into Winter, Spring, Summer, and Fall.

These additional features help compare flight delay performance, understand the contribution of different delay causes, and analyse seasonal patterns.

8. Analysis and Visualizations

- The flight dataset was analysed using Python to identify delay patterns, airport and airline performance, major causes of delays, and flight disruptions.
- The analysis was divided into five objectives to understand different aspects of flight delays across U.S. airports and airlines from 2020 to 2026.

9. Exploratory Data Analysis (EDA)

Analysis Performed:

1. Displayed the first five and last five records using `head()` and `tail()`.
2. Checked dataset dimensions using `shape`.
3. Examined column names using `columns`.

4. Checked data types and non-null values using `info()`.
5. Generated descriptive statistics using `describe()`.
6. Identified duplicate records using `duplicated()`.
7. Checked missing values using `isnull().sum()`.

Interpretation:

EDA helped understand the dataset structure, identify missing values and duplicates, and determine the data quality before performing further analysis.

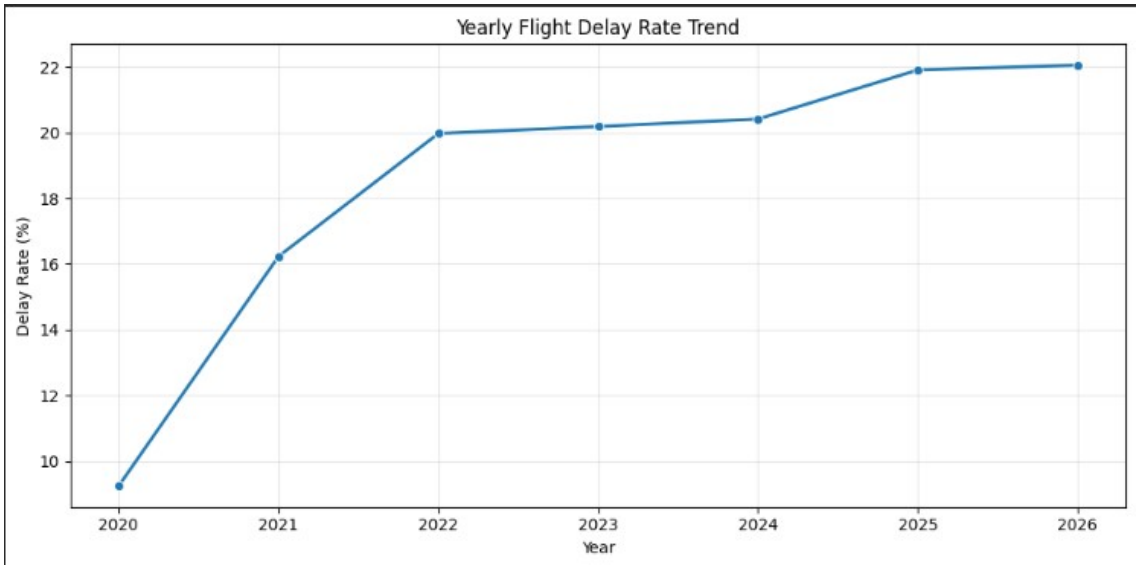
10. Data Visualization

Objective 1 – Analyse Flight Delay Trends Across Years and Months

Visualization 1: Yearly Flight Delay Rate

To analyse yearly flight delay rates from 2020 to 2026 and identify changes in flight delay patterns over time.

Visualization:



Interpretation:

- The lowest flight delay rate was recorded in 2020 at **9.24%**.
- The highest delay rate was recorded in 2026 at **22.06%**.
- The delay rate increased by **12.82 percentage points** between 2020 and 2026.
- The overall trend indicates an increase in flight delay rates over the recorded years.

Key Insight:

Flight delay rates show a significant increasing trend, more than doubling from 9.24% in 2020 to 22.06% in 2026.

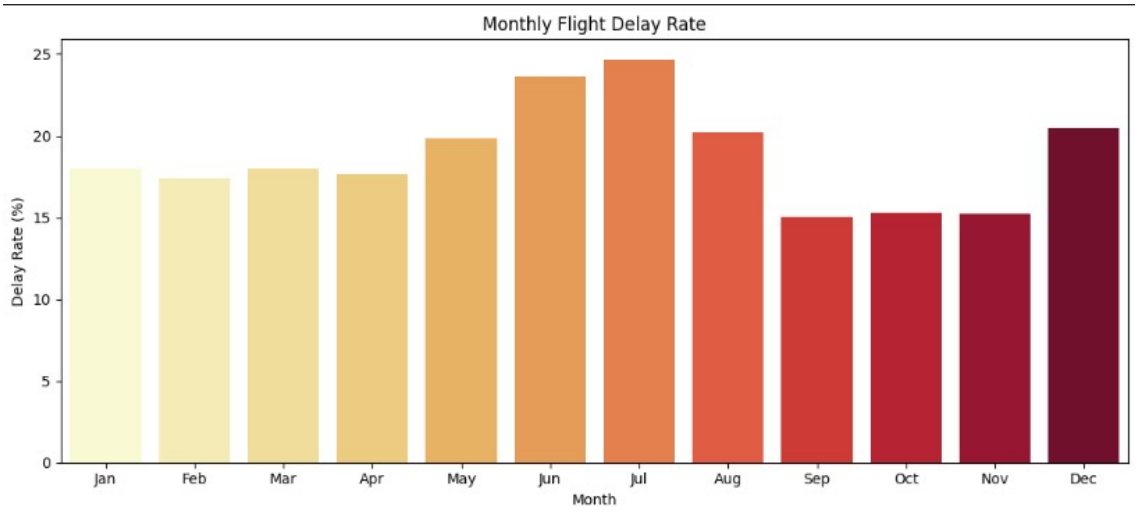
Business Insight:

Airlines should investigate the factors contributing to increasing delay rates and improve flight scheduling, operational planning, and resource allocation to reduce delays.

Visualization 2: Monthly Flight Delay Rate

To analyse flight delay rates across different months and identify the months with the highest and lowest delay rates.

Visualization:



Interpretation:

- July recorded the highest flight delay rate at **24.67%**.
- September recorded the lowest flight delay rate at **15.03%**.
- The difference between the highest and lowest monthly delay rates was **9.64 percentage points**.
- The results indicate that flight delays vary across months, with July experiencing considerably higher delays than September.

Key Insight:

July is the most delay-prone month, while September has the lowest delay rate among the months analysed.

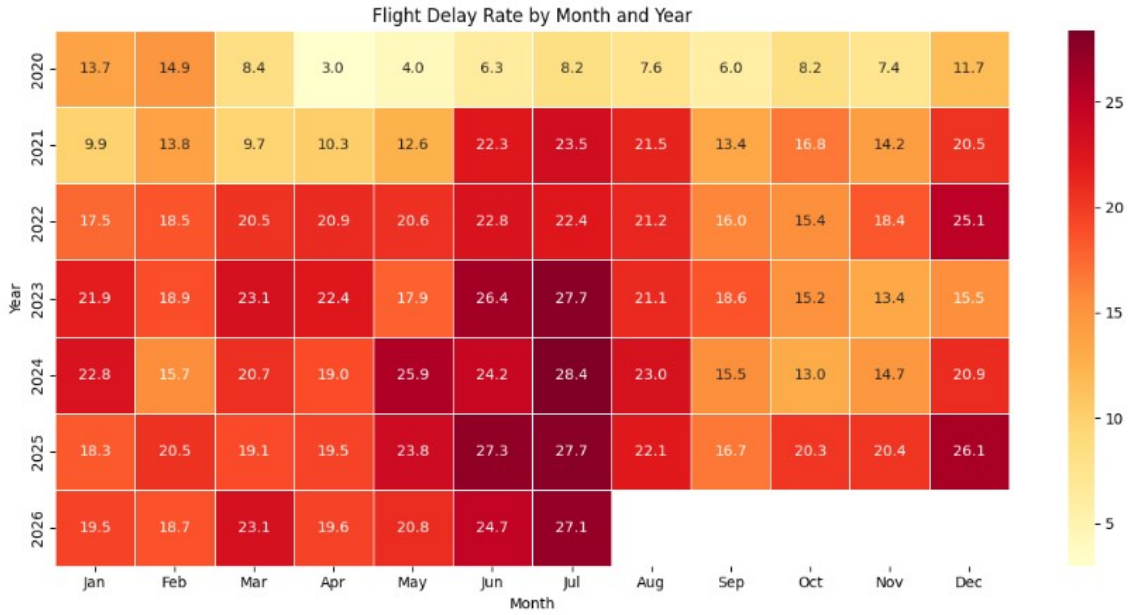
Business Insight:

Airlines can use these monthly patterns to plan flight schedules, allocate staff and aircraft resources more effectively, and prepare for periods with higher delay rates.

Visualization 3: Flight Delay Rate by Month and Year (Heatmap)

To analyse how flight delay rates vary across different months and years and identify specific periods with unusually high or low delay rates.

Visualization:



Interpretation:

- July 2024 recorded the highest flight delay rate at **28.37%** among the available month-year combinations.
- April 2020 recorded the lowest flight delay rate at **2.98%**.
- The difference between these two values was **25.39 percentage points**.
- The heatmap shows variations in flight delay rates across different months and years. Darker cells represent higher delay rates, while lighter cells represent lower delay rates.

Key Insight:

Flight delay rates vary considerably across different month-year combinations, with July 2024 experiencing the highest recorded rate.

Business Insight:

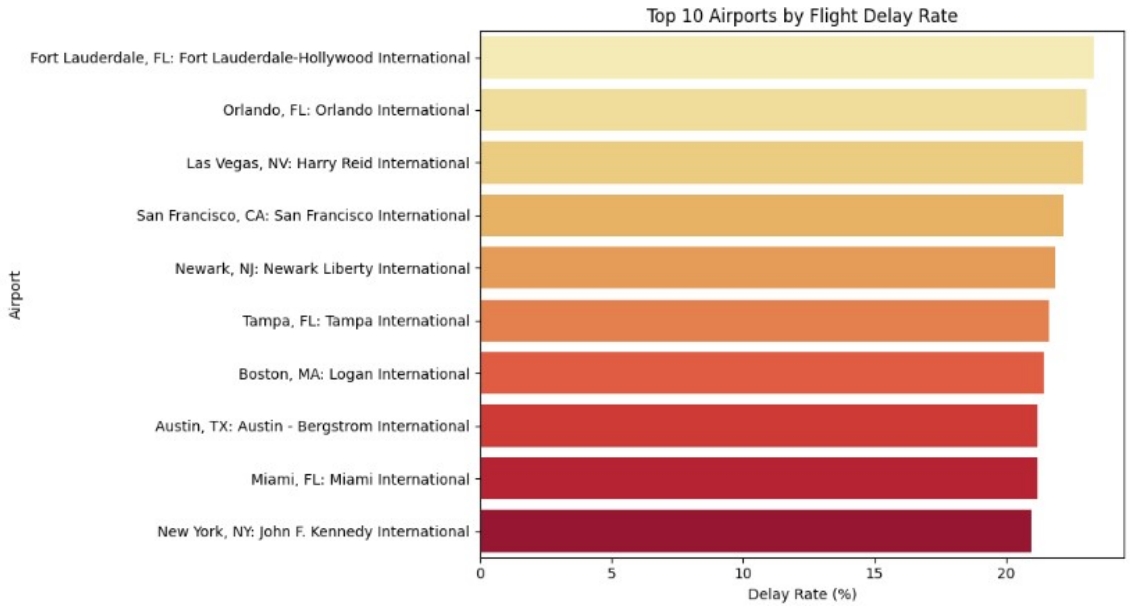
Airlines can use historical monthly and yearly delay patterns to identify periods of increased disruption, prepare additional operational resources, and improve seasonal flight scheduling.

Objective 2 – Analyse Airport and Airline Performance

Visualization 1: Top 10 Airports by Flight Delay Rate

To identify the 10 airports with the highest flight delay rates and compare their performance in terms of delayed flights relative to total flight operations.

Visualization:



Interpretation:

- Fort Lauderdale-Hollywood International Airport recorded the highest delay rate among the selected 10 airports at **23.31%**.
- The airport handled **560,569 flights**, of which **130,689 flights** were delayed.
- John F. Kennedy International Airport recorded the lowest delay rate among the selected 10 airports at **20.95%**.
- The difference between the highest and lowest airport delay rates was **2.36 percentage points**.

Key Insight:

Fort Lauderdale-Hollywood International Airport had the highest delay rate among the 10 selected airports, indicating comparatively lower on-time performance.

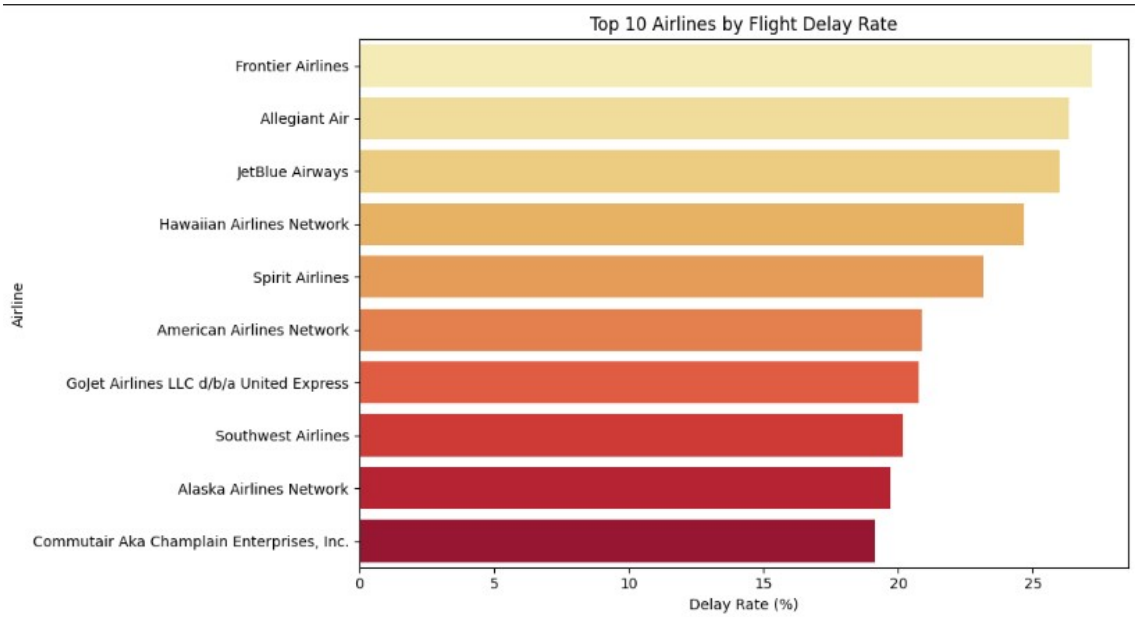
Business Insight:

Airports with higher delay rates should investigate possible operational bottlenecks, runway capacity constraints, and ground-handling efficiency. Airport management can use these findings to improve resource allocation and reduce flight disruptions.

Visualization 2: Top 10 Airlines by Flight Delay Rate

To identify the 10 airlines with the highest flight delay rates and compare their on-time performance based on the proportion of delayed flights.

Visualization:



Interpretation:

- Frontier Airlines recorded the highest delay rate among the selected 10 airlines at **27.19%**.
- It operated **816,057 flights**, of which **221,849 flights** were delayed.
- Commutair Aka Champlain Enterprises, Inc. recorded the lowest delay rate among the selected airlines at **19.12%**.
- The difference between the highest and lowest airline delay rates was **8.06 percentage points**.

Key Insight:

Frontier Airlines experienced the highest delay rate among the 10 selected airlines, indicating comparatively lower on-time performance.

Business Insight:

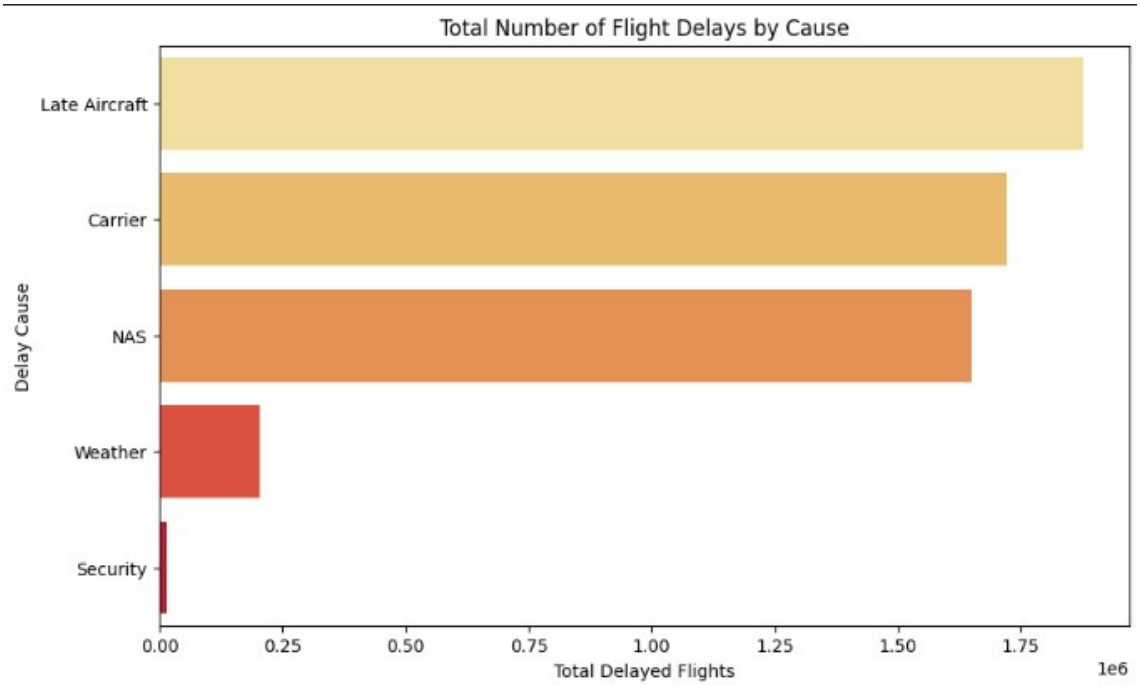
Airlines with higher delay rates should review their flight scheduling, turnaround procedures, crew availabili

Objective 3 – Identify the Major Causes of Flight Delays

Visualization 1: Total Number of Flight Delays by Cause

To identify the major causes of flight delays by comparing the total number of delays attributed to different causes.

Visualization:



Interpretation:

- Late Aircraft recorded the highest number of delays, with **1,877,404 delays**.
- Carrier-related issues recorded the second-highest number, with **1,721,544 delays**.
- NAS-related issues contributed **1,650,102 delays**.
- Weather-related issues accounted for **203,032 delays**.
- Security-related issues recorded the lowest number, with **14,754 delays**.

Key Insight:

- Late Aircraft was the leading contributor to the number of flight delays.
- Carrier and NAS were also major contributors.
- Security accounted for the fewest delays.

Business Insight:

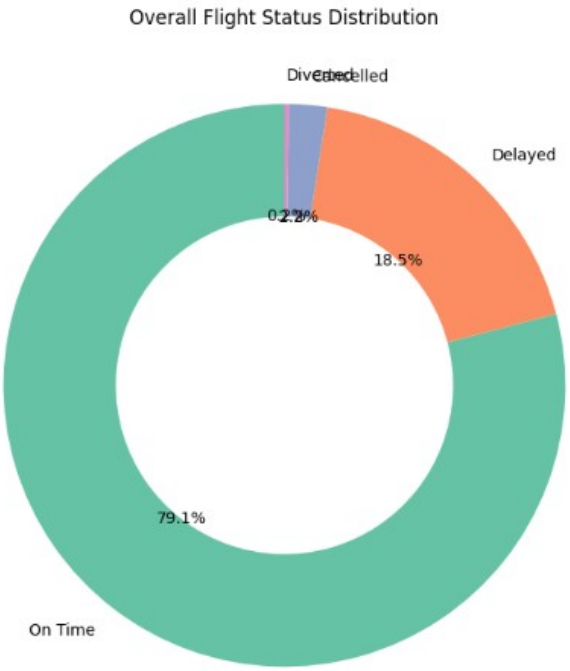
Airlines should focus on managing late-arriving aircraft, improving aircraft turnaround procedures, and investigating carrier-related and NAS operational issues. Addressing these major causes may help reduce flight delays and improve overall operational efficiency.

Objective 4 – Analyse Delayed, Cancelled, and Diverted Flights

Visualization 1: Distribution of Flight Status

To analyse the distribution of on-time, delayed, cancelled, and diverted flights and understand the proportion of flights affected by operational disruptions.

Visualization:



Interpretation:

- A total of **23,413,679 flights (81.07%)** operated on time.
- **5,466,880 flights (18.93%)** experienced delays of 15 minutes or more.
- **651,041 flights (2.25%)** were cancelled.
- **72,111 flights (0.25%)** were diverted.
- On-time flights accounted for the largest category, while diverted flights represented the smallest category.

Key Insight:

- Although 81.07% of flights operated on time, 18.93% experienced delays of 15 minutes or more, showing that flight delays remain a significant operational challenge.

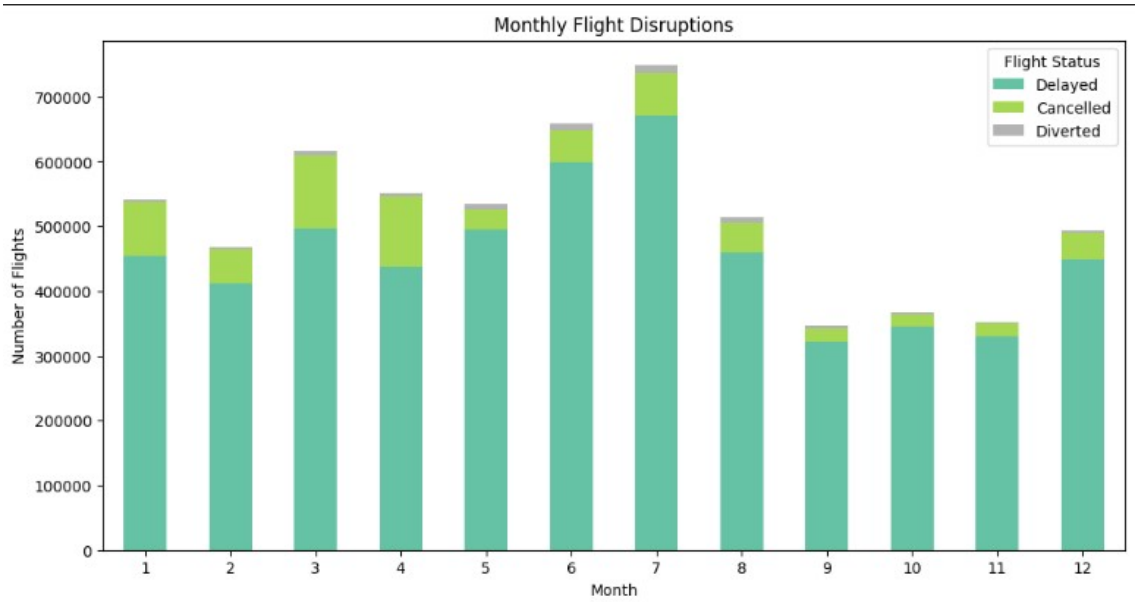
Business Insight:

Airlines can use these findings to monitor operational performance, improve punctuality, and develop strategies to reduce delays and cancellations.

Visualization 2: Monthly Distribution of Delayed, Cancelled, and Diverted Flights

To examine how delayed, cancelled, and diverted flight counts vary across different months and identify the months experiencing the highest operational disruptions.

Visualization:



Interpretation:

- July recorded the highest number of delayed flights, with **671,011 flights**.
- March recorded the highest number of cancelled flights, with **113,118 flights**.
- July also recorded the highest number of diverted flights, with **12,631 flights**.
- The results show that July experienced the highest delayed and diverted flight counts, while March had the highest cancellation count.

Key Insight:

- July was a particularly challenging month for flight operations, recording the highest numbers of delayed and diverted flights.

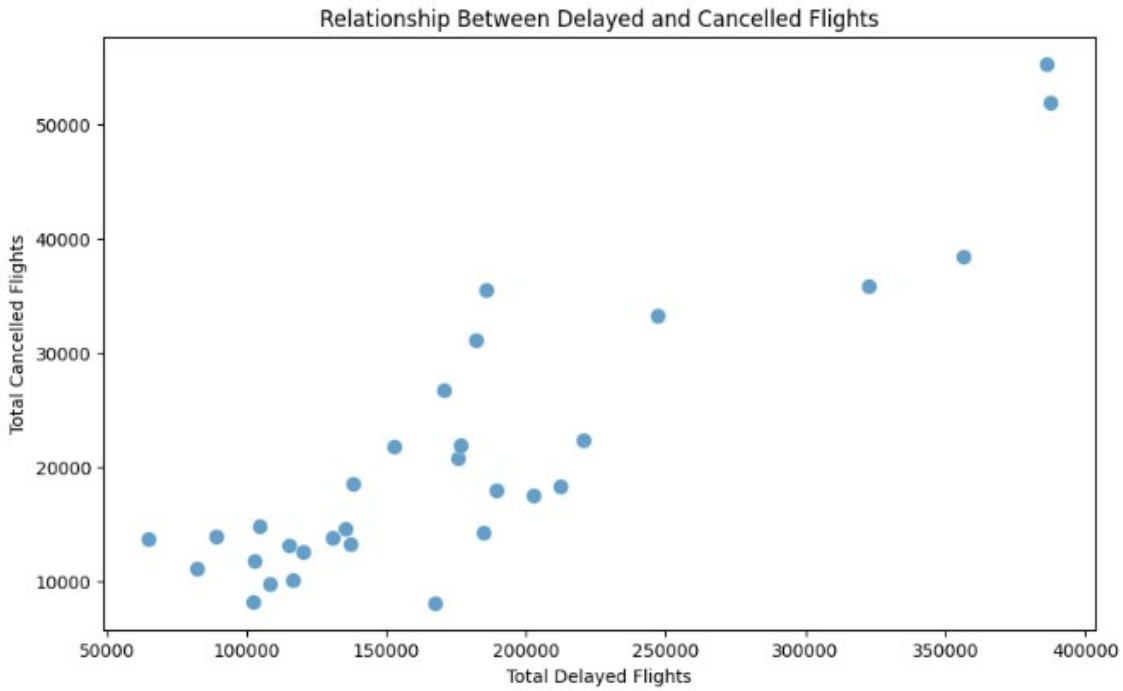
Business Insight:

Airlines should prepare for increased operational disruptions during July by improving flight scheduling and ensuring sufficient aircraft and crew availability. March may require additional attention to cancellation management.

Visualization 3: Relationship Between Delayed and Cancelled Flights Across Airports

To examine the relationship between delayed and cancelled flights across different airports and identify airports experiencing higher levels of flight disruptions.

Visualization:



Interpretation:

- The correlation between delayed and cancelled flights across airports was **0.89**, indicating a strong positive relationship.
- Dallas/Fort Worth International Airport recorded the highest number of cancelled flights among the airports analysed.
- The strong positive correlation indicates that airports with higher delayed-flight counts generally also tend to have higher cancelled-flight counts.

Key Insight:

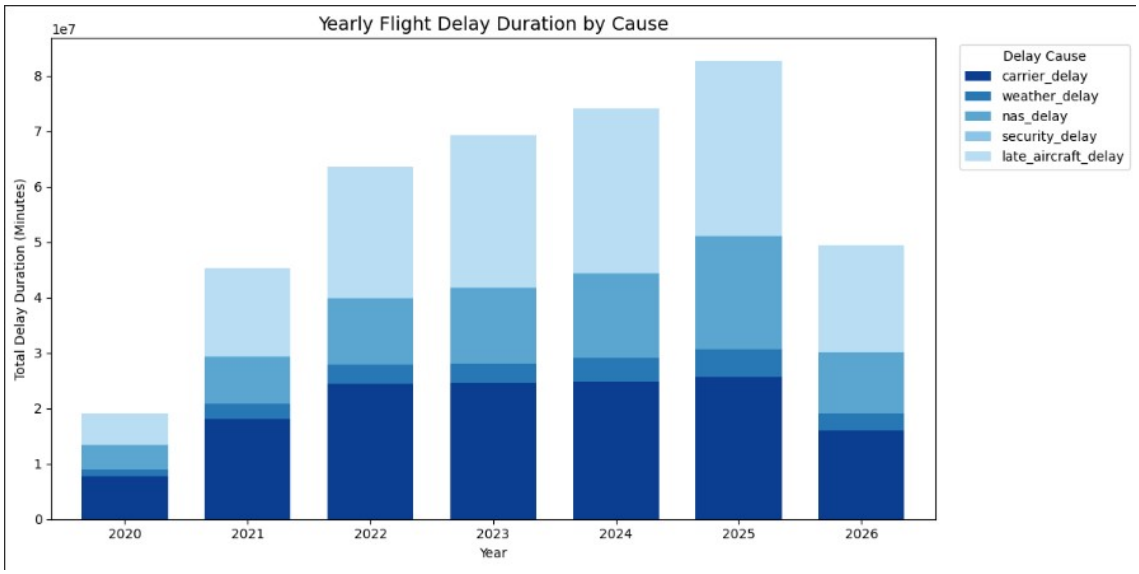
- Delayed and cancelled flights have a strong positive relationship, with a correlation coefficient of 0.89.

Business Insight:

Airport and airline management teams can use these findings to identify airports experiencing frequent operational disruptions. They can investigate congestion, scheduling issues, and resource constraints to improve operational efficiency and reduce cancellations.

Objective 5: To analyse total flight delay duration across different causes.

Visualization: Yearly Flight Delay Duration by Cause (Stacked Bar Chart)



Interpretation:

- 2025 recorded the highest total delay duration at 82.65 million minutes.
- Carrier delays were the leading cause from 2020 to 2022.
- Late Aircraft delays became the leading cause from 2023 onwards, reaching 31.52 million minutes in 2025.

Key Insights:

- Total delay duration increased between 2020 and 2025.
- Late Aircraft delays became the major contributor from 2023.
- Four out of five delay causes recorded their highest duration in 2025.

Business Insights:

- Airlines should focus on improving aircraft turnaround and scheduling to reduce delays caused by late-arriving aircraft.
- Better airline operations and resource planning can help reduce overall delay duration.

11. Types of Analysis

Data analysis can be classified into four major types: Descriptive, Diagnostic, Predictive, and Prescriptive Analysis. These approaches help understand past performance, identify reasons behind problems, predict future outcomes, and suggest possible improvements.

13.1 Descriptive Analysis

Descriptive analysis focuses on understanding what happened in the past by summarising historical data.

In this project:

- Analysed yearly and monthly flight delay rates.
- Compared airport and airline delay performance.
- Examined the distribution of on-time, delayed, cancelled, and diverted flights.
- Identified the major causes of flight delays.

13.2 Diagnostic Analysis

In this project:

- Identified late aircraft, carrier, weather, NAS, and security as different delay causes.
- Examined variations in delay duration across different years.
- Analysed the relationship between delayed and cancelled flights across airports.
- Identified months with higher flight disruption counts.

13.3 Predictive Analysis

In this project:

- Historical flight delay patterns were analysed to understand changes over time.
- However, no dedicated forecasting or machine learning model was developed to predict future flight delays.

13.4 Prescriptive Analysis

In this project:

- Recommended improving flight scheduling during months with higher delay rates.
- Suggested better aircraft turnaround and operational planning.
- Recommended investigating airports and airlines with higher delay rates.
- Suggested improving resource allocation to reduce cancellations and delays.

11. Key Findings

1. **Flight Delay Trends:** The overall flight delay rate increased from 9.24% in 2020 to 22.06% in 2026. July recorded the highest monthly delay rate at 24.67%.
2. **Airport and Airline Performance:** Fort Lauderdale-Hollywood International Airport had the highest delay rate among the selected airports at 23.31%. Frontier Airlines recorded the highest delay rate among the selected airlines at 27.19%.
3. **Major Causes of Delays:** Late Aircraft was the leading cause by number of delays, contributing 1,877,404 delays, followed by Carrier and NAS-related issues.
4. **Flight Disruptions:** Delayed flights accounted for 18.93% of the flight records analysed. A strong positive correlation of 0.89 was observed between delayed and cancelled flights across airports.
5. **Delay Duration:** Late Aircraft contributed the highest total delay duration, with 153,076,879 minutes, making it the largest contributor among the five delay causes.

12. Business Insights

1. **Improve Flight Scheduling:** Since flight delay rates increased over the analysed years, airlines should regularly review their flight schedules and identify periods with higher delays. Additional time between connecting flights may help reduce operational disruptions.
2. **Improve Airport and Airline Operations:** Airports and airlines with higher delay rates should review their operational processes, identify the reasons for delays, and take steps to improve efficiency.
3. **Manage Major Delay Causes:** Late Aircraft, Carrier, and NAS-related issues contribute to the highest number of delays. Airlines should focus on improving aircraft turnaround time, maintenance planning, crew scheduling, and coordination with air traffic management.
4. **Reduce Flight Disruptions:** The positive relationship between delayed and cancelled flights highlights the importance of effective disruption management. Airlines should improve contingency planning and provide timely updates to passengers.
5. **Focus on Reducing Delay Duration:** Since late-arriving aircraft contribute the highest total delay duration, airlines should improve aircraft turnaround procedures and scheduling to minimise delays and their impact on subsequent flights.

13. Conclusion

This project analysed U.S. aviation bottlenecks using Python, Pandas, Matplotlib, and Seaborn. It examined flight delay trends, airport and airline performance, major delay causes, and flight disruptions.

The analysis identified Late Aircraft as the leading contributor to flight delays. Overall, the project provided insights into aviation bottlenecks and demonstrated how data analysis can support better operational decisions.